DG INTERNS HUB — TECHNICAL NOTES

TOPIC 06: WAZUH ARCHITECTURE & CORE COMPONENTS

Submitted By: Gunjan
Department: Cybersecurity / SOC
Internship: DG Interns Hub Cybersecurity
Week: Week 4
Submission Date: 31 October 2026
Document ID: 06_Wazuh_Architecture.pdf

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1. Core Component Breakdown

- **Wazuh Agent:** Installed on target endpoints. Performs log collection, rootkit detection, vulnerability scanning, and File Integrity Monitoring (FIM).
- **Wazuh Manager:** Central analysis node that receives encrypted events over port 1514, decodes raw strings via regex patterns, and triggers XML correlation rules.
- **Wazuh Indexer:** High-performance OpenSearch node that indexes and stores alert documents.
- **Wazuh Dashboard:** Kibana-based user interface for data exploration, alert triage, and rule management.

2. Wazuh Rule Engine Example

```
<!-- Custom Rule for SSH Brute Force Detection in Wazuh -->
<group name="syslog,sshd,">
  <rule id="100050" level="10">
    <if_matched_sid>5716</if_matched_sid>
    <same_source_ip />
    <frequency>8</frequency>
    <timeframe>120</timeframe>
    <description>Possible SSH Brute Force Attack Triggered</description>
    <mitre><id>T1110</id></mitre>
  </rule>
</group>
```

3. Practical SOC Scenario: Active Response Remediation

Scenario: Automated IP Block via Active Response

Incident Context: External host `203.0.113.88` attempts rapid SSH logins into a Linux Bastion host.

Detection & Response: Rule 100050 triggers. Wazuh Manager instructs the agent via `active-response/bin/host-deny` script to automatically append `203.0.113.88` to `/etc/hosts.deny` for 30 minutes.

SOC Outcome: Attack automated mitigation occurs in real-time without requiring manual analyst intervention.

4. Conclusion

Wazuh combines log collection, rule evaluation, and active threat response into a single unified architecture, providing robust open-source endpoint protection and SIEM capabilities.